Detecting and Quantifying Airline Delay Cascades at Scale Using PySpark RDDs

Midterm progress report (45% stage)

STA 473, Tools and Techniques for Data Science | 10 November 2026

Submitted to: Sir Farooq

Team member	Registration no.
Ahmad Fayyaz	SP24-BST-002
Hajira Shahbaz	SP24-BST-010

1. The project in simple words

The idea in 30 seconds. When a plane lands late, its next flight usually leaves late too, and so does the one after that. One late flight can spoil the whole day for that plane, like dominoes falling one after another. We call this a **delay cascade**. Airlines can see that a flight was late, but they cannot easily see **which flight started the problem** or **how much extra delay it caused later**. Our project finds the starting flight and counts the delay it caused.

What we do. We take about 5.8 million real US flights and use PySpark (a tool for processing very large data) to rebuild each plane's day, flight by flight. Then we find the first flight where each delay started, called the **root flight**, and add up the delay minutes it caused later. That total is the **Cascade Impact Score**. With it, an airline can see which flights, airports and routes spread the most delay, and where giving planes more time on the ground would stop the spreading.

What is new. Many projects try to predict whether a flight will be late. We do something different: we measure how much damage a late flight causes afterwards, and we test which change in the schedule would reduce it.

The questions we want to answer

- How much of all delay comes from a plane arriving late from its previous flight?
- How long are the delay chains, and does this change during the day?
- Which airports and routes spread delay the most?
- Does more time on the ground between flights stop a delay from spreading?

2. Data

We use the public Kaggle dataset "2015 Flight Delays and Cancellations" (flights.csv, airlines.csv, airports.csv). It has about 5.8 million US domestic flights from the whole of 2015. The original source is the **US Bureau of Transportation Statistics (BTS)**, a US government office that collects official transport data. Every airline reports each domestic flight to BTS: the plane's ID (tail number), planned and actual times, delay minutes, and the reason for the delay. The data is free to use (CC0 licence). It is large enough for both the midterm and the final report.

The data also records **why** a flight was late (airline, weather, air traffic system, security, or late aircraft). We use this to check our results later.

3. How the pipeline works

The work runs in Google Colab with PySpark. Steps 1 to 4 (cleaning) are finished and the cleaned data is saved. Steps 5 to 8 (finding cascades) are working as a prototype on January and February.

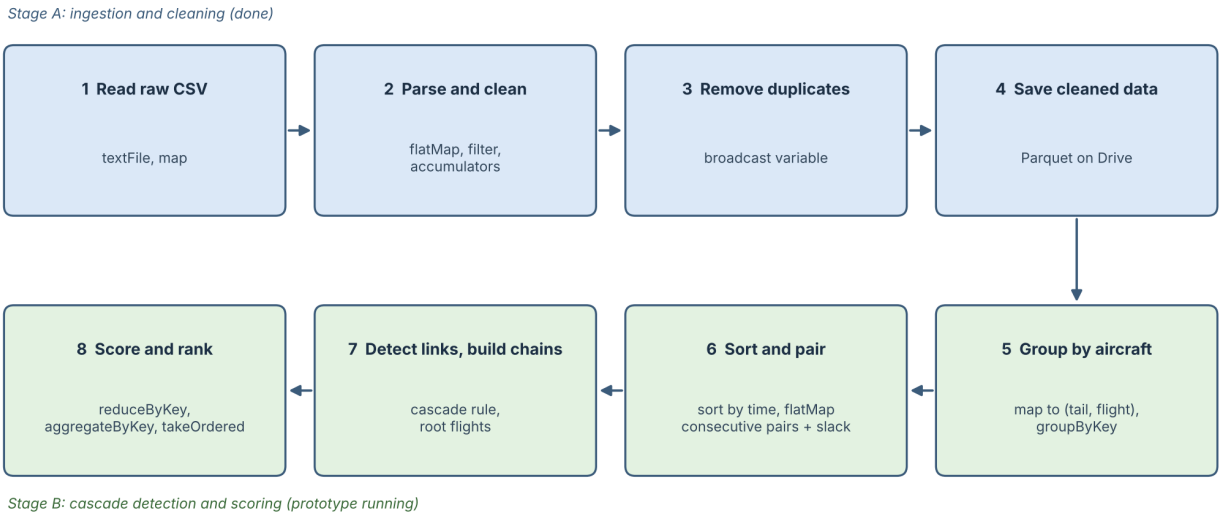


Figure 1. The pipeline from raw file to final ranking.

RDD means Resilient Distributed Dataset: a big list of data split into pieces so that many workers can process the pieces at the same time. The table shows which RDD tools we use and for what.

RDD tool	What we use it for
textFile, map, flatMap	Read the big file line by line and turn each line into a flight record
filter	Throw away cancelled, diverted and broken rows
Accumulators	Counters that all workers add to, used to count how many rows we removed and why
Broadcast variable	A small list sent to every worker, used to find and remove 1,194 duplicate rows
groupByKey	Put all flights of the same plane together, so we can see its day
reduceByKey, aggregateByKey	Add up values per group, such as total delay per airport or carrier
cache	Keep data in memory so repeated work is faster
takeOrdered, sortBy	Pick the top results, such as the worst airports or chains

4. Cleaning the data and what we found

Real data is messy, so we cleaned it first using RDDs. We counted every removed row with an accumulator, and the numbers add up exactly: rows read minus rows removed equals rows kept.

Step	Rows
Rows read	5,819,079
Broken or unreadable rows	0
Cancelled or diverted flights removed	105,071
Rows with no tail number	0 (cancelled flights usually have none and were already removed)
Duplicate rows removed (same plane, same departure minute)	1,194
Clean flights kept	**5,712,814**
October rows with numeric airport codes (flagged, kept)	482,878

Things we found in the data

- **October airport codes.** In October, airports are written as 5-digit numbers (for example 10155) instead of 3-letter codes like ORD. This affects 482,878 rows. We can still follow planes and delays in October, because the same airport always has the same number. But we cannot name airports for October yet. We flagged these rows instead of deleting them. Our plan is to match them with the same flights in September or November to recover the codes, or to leave October out of airport rankings only.
- **Late-aircraft reason.** The data never gives this reason for flights less than 15 minutes late. This fits our choice of 15 minutes as the starting point for a cascade.
- **Time zones.** Flight times are in local time at each airport. We checked all flights and the time-zone gap was always between -6 and +6 hours, which is normal for the US. A small number of flights (14,556, or 0.25%) seem to land before they take off on the clock. These are short flights going west across a time zone, so we kept them.
- **Cancelled and diverted flights** (about 1.8%) have no delay minutes, so we removed them. This can leave a hole in a plane's day, so we link two flights only if the first lands where the second takes off. We list this as a limit.
- **Extreme values.** 266 flights arrived more than 1,000 minutes late and 392 arrived more than an hour early. These are real, so we kept them and report them.

5. How we find a cascade and score it

For each plane, we put its flights in time order and look at every pair of neighbouring flights: flight A, then the next flight B. The **spare time** (called slack in the research papers) is the planned time on the ground between A landing and B taking off. Airlines plan this time to absorb small delays. We say A passed delay on to B when all four things are true:

- A landed at least 15 minutes late (we also test other values);
- the planned spare time was short, at most 60 minutes (we also test 45 and 90);
- B left late, and the data gives "late aircraft" as part of its reason;
- A's lateness was bigger than the spare time, so the ground time could not absorb it.

The fourth rule matches the definition in a research paper by Brueckner and co-authors (2022): a delay spreads only when it is bigger than the buffer. The amount that really passes on is:

> **Delay passed on = the smaller of: (A's lateness minus the spare time) and (how late B left)**

Root flight. A flight that passes delay on but did not receive any itself. Following the passed-on delay from the root, flight after flight, gives the **chain**.

Cascade Impact Score. The total minutes of delay passed on along the whole chain, counted from the root flight. We also compute a **spread ratio**: the score divided by the root flight's own delay. A ratio above 1 does not mean the delay got bigger. It means one delay was carried through several flights, and the minutes of all those flights add up. We then add up scores by airport, route, hour and airline.

A fix we made while testing. At first we counted a pair as passing delay even when the spare time was bigger than the lateness. This gave chains with a score of 0. We fixed the rule so that delay counts as passed on only if it was really bigger than the spare time. We threw away the numbers from before the fix.

6. Real examples of cascades (prototype results)

We ran the detector on January and February 2015 (864,516 flights). These three chains are real and were printed by our notebook after the fix. All delays and spare times are in minutes.

Example A: the worst case (plane N928FR, airline F9, 7 January). Cascade Impact Score: 1,304 minutes

Leg	Flight	Route	Planned takeoff	Planned landing	Late at takeoff	Late at landing	Spare time	Passed on
1	1279	PHL-MCO	05:30	08:10	301	321	40	281 (root)
2	1278	MCO-PHL	08:50	11:10	321	337	40	297
3	1281	PHL-TPA	11:50	14:25	372	406	40	366
4	1280	TPA-PHL	15:05	17:30	409	400	40	360
5	1277	PHL-MCO	18:10	20:50	394	477	-	-

The first flight left Philadelphia 301 minutes late. All day, every flight of this plane stayed 300 to 400 minutes late. The 40 minutes of spare time did not help, because the delay was much bigger than that.

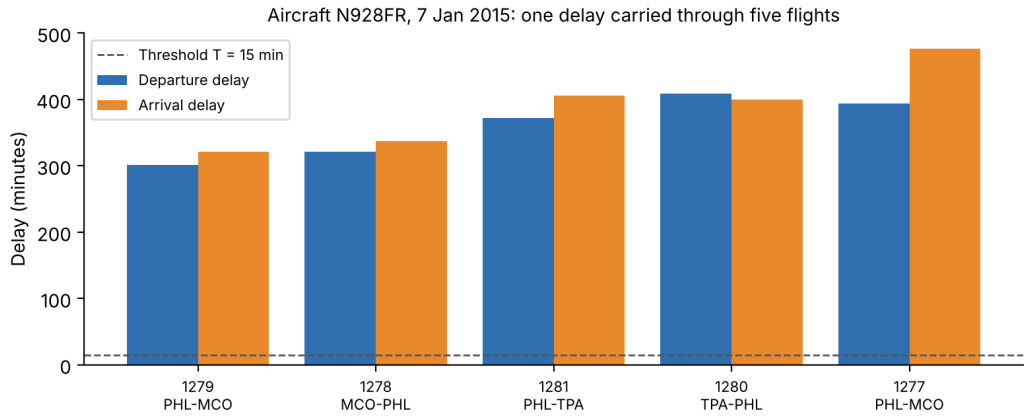


Figure 2. Lateness at each flight of the worst chain.

Example B: a typical chain (plane N710EV, airline EV, 3 February). Cascade Impact Score: 33 minutes

Leg	Flight	Route	Planned takeoff	Planned landing	Late at takeoff	Late at landing	Spare time	Passed on
1	5477	DTW-ELM	10:00	11:11	39	55	35	20 (root)
2	5477	ELM-DTW	11:46	13:09	49	44	36	8
3	5348	DTW-BUF	13:45	15:00	62	40	35	5
4	5348	BUF-DTW	15:35	16:53	37	21	-	-

Flight 1 landed 55 minutes late and the plane had 35 minutes of spare time, so 20 minutes were passed on. Flight 2 passed on 8 and flight 3 passed on 5. The score is 20 + 8 + 5 = 33. Notice that flight 3 left 62 minutes late but only 8 of those minutes came from the previous flight. The rest was its own delay. That is why we count only passed-on minutes.

Example C: a short chain (plane N615MQ, airline MQ, 24 February). Cascade Impact Score: 5 minutes

Leg	Flight	Route	Planned takeoff	Planned landing	Late at takeoff	Late at landing	Spare time	Passed on
1	3420	ORD-BDL	10:15	13:18	36	32	27	5 (root)
2	3319	BDL-ORD	13:45	15:31	34	46	-	-

Short chains like this are the most common kind. The spare time absorbed most of the delay and only 5 of the 32 minutes were passed on.

Summary for all of January and February

Out of 864,516 flights, the detector looked at 839,235 pairs of back-to-back flights of the same plane. Only **37,143 (4.4%)** were real delay-passing links. Joined together, they formed **24,349 chains**.

Chain length (flights)	2	3	4	5	6	7	8
Number of chains	16,498	4,617	2,033	808	295	81	17
Share of all chains	67.8%	19.0%	8.4%	3.3%	1.2%	0.3%	0.1%

Two out of three chains are short (one link only). About one chain in three has three or more flights, and the longest ones reach 8 flights.

The five chains with the highest Cascade Impact Score

Plane	Root flight	Root landed late (min)	Chain length	Score (min)	Spread ratio
N928FR	1279 PHL-MCO	321	5	1,304	4.06
N210FR	1256 ORD-MIA	686	3	1,275	1.86
N910AT	2673 BOS-LGA	130	6	944	7.26
N755EV	4950 MSP-MDW	205	7	937	4.57
N941FR	1046 MCO-MDT	333	4	928	2.79

The worst chain (N928FR) is Example A above. Notice that the highest scores come from planes that were already very late at the start and had too little spare time to recover, not only from long chains.

Airports where the most delay is passed on

Rank	Airport	Minutes passed on	Rank	Airport	Minutes passed on
1	ORD (Chicago O'Hare)	79,156	6	MCO (Orlando)	40,704
2	LGA (New York LaGuardia)	63,061	7	LAS (Las Vegas)	39,880
3	DEN (Denver)	57,943	8	LAX (Los Angeles)	38,299
4	SFO (San Francisco)	51,327	9	PHX (Phoenix)	32,989
5	ATL (Atlanta)	45,722	10	BOS (Boston)	32,333

These are total passed-on minutes at the airport where the plane was waiting between two flights. Busy airports have more flights, so their totals are higher. In the next stage we will also compute a rate per flight, so that busy and quiet airports can be compared fairly.

7. What the data shows (full year, 5,712,814 flights)

We used RDD tools (aggregateByKey and reduceByKey) to calculate average arrival delay, in minutes, over the full year. For airports we used the departure airport, only airports with at least 10,000 flights, and left out the October rows with numeric airport codes.

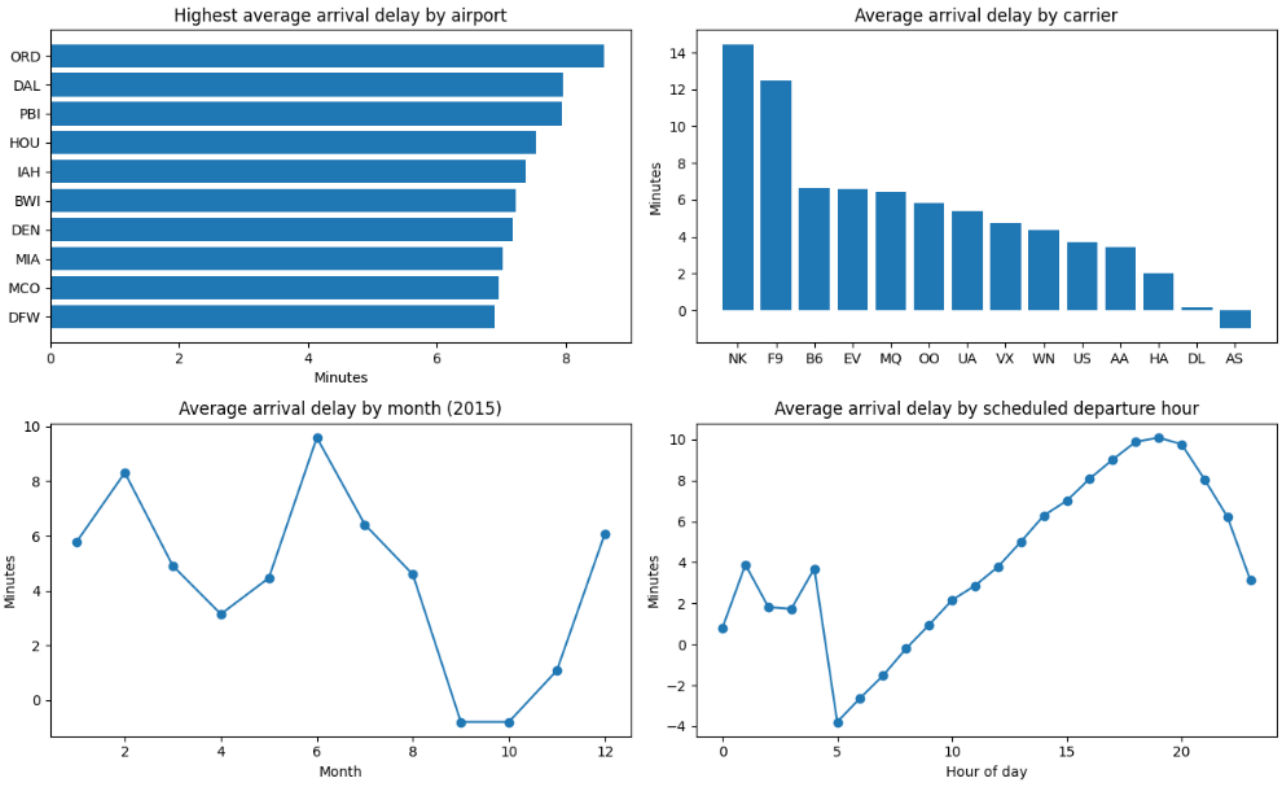


Figure 3. Average arrival delay by airport, airline, month and planned departure hour.

- **Time of day.** Early flights (05:00 hour) arrive slightly early on average (-3.79 min). Delay then grows almost steadily through the day and peaks at 10.09 min for flights leaving in the 19:00 hour, before falling to 3.12 min at 23:00. This fits the idea that late planes carry their delay into later flights, so delay builds up during the day. Our chain analysis will test this directly. Hours 0 to 4 have very few flights (526 to 14,504 each), so those averages are not reliable.

- **Airlines.** Spirit (NK, 14.47 min) and Frontier (F9, 12.48 min) are far worse than the rest. All other airlines are between Alaska (AS, -0.98 min) and JetBlue (B6, 6.67 min), and Delta (DL) is 0.17 min. Frontier is also the airline of our worst example chain. Low-cost airlines plan very short ground times, which research links to more spreading of delay.
- **Months.** June (9.59 min) and February (8.31 min) are the worst. September and October are the best (-0.79 min). We have not tested why, but the weather column in the data can show whether winter and summer weather explain it.
- **Airports.** Chicago O'Hare (ORD, 8.58 min, 276,494 flights) is highest, then DAL (7.95), PBI (7.94), HOU (7.54), IAH (7.37), BWI (7.22), DEN (7.18), MIA (7.02), MCO (6.95) and DFW (6.90).

Why were flights late? (our first research question)

Of the 5,712,814 clean flights, 1,062,804 (18.6%) landed 15 minutes or more late. For these, BTS records a reason. **A late incoming aircraft is the biggest single reason, at 39.8% of all delay minutes.**

Reason	Delay minutes	Share
Late aircraft (previous flight was late)	24,926,341	39.8%
Airline (crew, maintenance, cleaning and so on)	20,123,433	32.2%
Air system (air traffic control, airport congestion)	14,330,316	22.9%
Weather	3,097,537	5.0%
Security	80,974	0.1%

The airlines report these reasons themselves. Our chain analysis measures spreading separately, from the plane sequences, and we will compare the two results.

8. What other researchers have done (literature review summary)

Two kinds of delay. A *primary delay* has its own cause, such as bad weather or a technical fault. A *reactionary* (or knock-on) delay happens only because an earlier flight was late (EUROCONTROL, 2010). A joint study by EUROCONTROL and RWTH Aachen University found that 40% to 50% of delays in Europe were reactionary, and that hub-and-spoke operations had less of it than point-to-point and low-cost flying. In the US data, the "late aircraft" reason is the same idea.

Spare time on the ground. Brueckner, Czerny and Gaggero (2022) say a delay spreads when the incoming lateness is bigger than the planned buffer before the next flight. They look only at a flight and the one just before it. We use the same buffer idea, but we follow the whole chain. AhmadBeygi and co-authors (2008) show that schedules built to use planes as much as possible leave little spare time, so delays spread more. In a later paper (2010) they show how moving existing spare time around can reduce spreading. These papers support our test of how much spare time helps.

The whole network. Fleurquin, Ramasco and Eguiluz (2013) built a computer model of the US network using 2010 flight data. They found delay spreads in three ways: through the same plane, through passengers and crew connecting between flights, and through airport congestion. Passenger and crew connections had the strongest effect. Pyrgiotis, Malone and Odoni (2013) used a queuing model of the 34 busiest US airports. Both are models. We work directly from recorded delays, but we follow only the plane's own sequence.

Bottleneck flights. Arikan, Deshpande and Sohoni (2013) estimate how a starting delay spreads along a plane's day and define a measure for "bottleneck" flights. This is close to our root flight and score.

Their model estimates what is expected to happen, while our score counts the minutes that really passed down each real chain.

What is different in our project. Among the papers we read: (1) we rebuild the full chain of each plane from the data, not just one flight and its neighbour; (2) we score the damage each delay causes, for flights, airports and routes; (3) we process about 5.7 million flights with a PySpark RDD pipeline and will compare it with DataFrames; and (4) we check our results against the reasons in the data and against made-up cascades where we know the right answer. The full literature review is a separate document.

9. Real-world use and what problem it solves

The problem. Airlines know delays happen, but not which delays do the most damage later in the day. So they often fix symptoms instead of causes. Our score ranks flights, airports, routes and times of day by the delay they cause downstream, and the spare-time analysis shows where extra ground time would stop a cascade.

How it could be used. Our project works on past data (a batch analysis). In real life the same pipeline could run every night on the previous day's flights and give planners an updated ranking and advice on ground time. A live version would read flights as they happen, for example with Spark Structured Streaming. That live version is future work and is not built in this project.

Limits of our work

- We follow only the plane's own sequence. Crew, passenger and airport-crowding effects are not in the BTS data.
- Cancelled and diverted flights (about 1.8%) are removed, so a chain cannot continue across a missing flight.
- October airport codes are numbers, so October is left out of airport rankings for now.
- The 15-minute and 60-minute limits are our choices, so we test other values.
- The cascade results so far come from January and February only, and winter weather may affect them.

10. Who does what

Area	Ahmad Fayyaz (SP24-BST-002)	Hajira Shahbaz (SP24-BST-010)
Setup	Colab, GitHub repo, data download	Literature review, problem definition
Pipeline	RDD reading, cleaning, accumulators, saving Parquet	Exploratory RDD calculations and charts
Core	Building each plane's chain of flights	Cascade rules, limits, first Impact Score
Made-up data	Pakistan-style flight generator	Planted cascades with known answers
Later	Speed test, optimisation, what-if simulation	Rankings, checking results, testing limits, recommendations

Each of us reviews the other's code and can explain every part of the project.

11. Progress so far and the plan for the rest

Item for the 45% stage	Status
Topic, problem statement, objectives and proposal	Done
Literature review (3 to 5 sources)	Draft written (5 main sources)
Data downloaded and described	Done
RDD pipeline with accumulators and cleaning report	Done
Chain builder for each plane	Done (prototype on January and February)
Prototype detector with 2 to 3 real example chains	Done (Section 6)
First version of the Cascade Impact Score	Done
Exploratory results: by airport, airline, month, hour, and share of late-aircraft delay	Done (Section 7)
Made-up Pakistan-style data and first check	To do (design plus small sample)
Architecture diagram	Done (Figure 1)
Slides	To do

The remaining 55%

When	Work
Weeks 5 to 7	Run on the full year, impact score for flights, airports and routes, find which airports spread delay most
Weeks 8 to 9	RDD vs DataFrame speed test, check results, write recommendations
Week 10	Final report and presentation

Planned extras. Must do: a what-if test of adding more ground time, a graph of spare time against spreading, and a check on made-up cascades where we know the right answer. If time allows: testing different limits, a speed comparison of RDD and DataFrame, a network graph of cascades, and a comparison with weather.

Glossary: terms in simple words

Term	What it means
PySpark	A Python tool that processes very large data by splitting the work across many processors at once.
RDD	A big list of data cut into pieces so that many workers can process the pieces at the same time.
Accumulator	A counter that all workers can add to. We use it to count removed rows.
Broadcast variable	A small list sent to every worker so each one can look things up quickly.
groupByKey / reduceByKey	Put records with the same label together, or add up their values. For example: all flights of one plane, or total delay per airport.
Parquet	A compact file format for big tables, so we do not have to clean the data again.
Tail number	The ID of one physical aircraft, like a car's number plate.
Spare time (slack)	Planned time on the ground between landing and the next takeoff.
Root flight	The flight where a chain of delay starts.
Cascade Impact Score	Total minutes of delay that a root flight passed on to later flights.
Primary delay	A delay with its own cause, such as weather or a technical fault.
Reactionary (knock-on) delay	A delay that exists only because an earlier flight was late.
BTS	US Bureau of Transportation Statistics, the government office that publishes the flight data.

References

AhmadBeygi, S., Cohn, A., Guan, Y., & Belobaba, P. (2008). Analysis of the potential for delay propagation in passenger airline networks. *Journal of Air Transport Management, 14*(5), 221-236. https://doi.org/10.1016/j.jairtraman.2008.04.010

AhmadBeygi, S., Cohn, A., & Lapp, M. (2010). Decreasing airline delay propagation by re-allocating scheduled slack. *IISE Transactions, 42*(7), 478-489.

Arikan, M., Deshpande, V., & Sohoni, M. (2013). Building reliable air-travel infrastructure using empirical data and stochastic models of airline networks. *Operations Research, 61*(1), 45-64.

Brueckner, J. K., Czerny, A. I., & Gaggero, A. A. (2022). Airline delay propagation: A simple method for measuring its extent and determinants. *Transportation Research Part B: Methodological, 162*, 55-71. https://doi.org/10.1016/j.trb.2022.05.003

EUROCONTROL. (2010, May 10). *News release on the study of delay propagation in Europe, carried out with RWTH Aachen University.* https://www.airport-suppliers.com/supplier-press-release/eurocontrol-news-5

Fleurquin, P., Ramasco, J. J., & Eguiluz, V. M. (2013). Systemic delay propagation in the US airport network. *Scientific Reports, 3*, 1159. https://doi.org/10.1038/srep01159

Pyrgiotis, N., Malone, K. M., & Odoni, A. (2013). Modelling delay propagation within an airport network. *Transportation Research Part C: Emerging Technologies, 27*, 60-75. https://doi.org/10.1016/j.trc.2011.05.017

US Bureau of Transportation Statistics. Reporting Carrier On-Time Performance data (as published on Kaggle: 2015 Flight Delays and Cancellations, CC0).